

SEDAR 61 Gulf of Mexico Red Grouper

Age and Growth

- Investigate methods to better collect age structure samples randomly and systematically from all fishing sectors, especially the recreational sector which is highly under represented
- Explore growth model alternatives that includes both the non-random sampling due to minimum size restrictions (Diaz et al. 2004) and non-random sampling due to biases in over/under sampling specific length bins (Chih 2014a, 2014b).
- Continue collaboration with ageing facilities throughout the Gulf of Mexico and South Atlantic. These efforts will include the annual reading of references sets for Red Grouper and other reef fish, and annual meetings to review the interpretation of ageing structures and the timing of annual band deposition.
- Continue ongoing research evaluating the potential for aging errors (edge type definitions, quality control, seasonal trends, etc.) of Red Grouper (among other reef fish) discussed in SEDAR61-DW17 and SEDAR62-DW18 (posted for SEDAR62) to determine if and how age assignment problems could affect the estimation of both age frequency distributions and growth curves and whether alternative methods (e.g., using second season ALKs or length based assessment models) may be needed to address these potential issues.
- Explore the use of Fourier Transform Near-Infrared Spectroscopy (FT-NIRS) to derive ages for Red Grouper and other reef fish
- Ensure robust communication between age reading laboratories and stock assessment scientists to assure a mutual understanding of the age advancement protocols for age readers and the age advancement protocols used in the assessment models. Concerns raised could be further explored during subsequent SEDAR assessments for those species.

Discard Mortality

- Continue data collection from observer programs

Maturity/Sexual Transition

- Explore changes in reproductive parameters over time and space(e.g., Moe [1969]vs now)
- Explore choice of criteria to assign maturity

Fecundity

- Explore appropriate measures of reproductive potential such as combined male/female SSB which has been more commonly applied for protogynous fishes (Shepherd et al. 2013)

Landings

- Re-evaluate historical landings in light of the new MRIP estimates of catch and effort and revise as necessary
- Assign annual uncertainty estimates (e.g., SE) to historic and recent commercial and recreational landings by fishery, which would allow the assessment to include all available landings data while accounting for greater uncertainty in the historic period

Discards

- Obtain consistent funding source to ensure continuation of sampling of discard length composition for Red Grouper and other species

Commercial CPUE indices

- Additional research is needed to better understand the influence of the IFQ program on fisher behavior and investigate alternative analyses

Recreational CPUE indices

- Additional research is needed to investigate if assumptions are appropriate across full time series (e.g., targeting, trip length, effects of various regulations, red snapper)

Surveys

- Use of fishery-independent data, such as from the Combined Video Survey, to explore the spatial overlap of red tide with Red Grouper (and other reef fishes)

MRIP size data

- Conduct a simulation study to evaluate whether different imputation processes (e.g., different imputation methods, algorithms, validity of assumptions, etc.) actually produce benefits that outweigh the uncertainties of adding imputed data to the observed data
- Determine the feasibility of developing weighting factors for data sources other than MRIP or how to determine effective sample sizes when combining various data sources

Composition data

- Consider using the number of stations or trips from which the compositions came as input sample sizes for composition data, rather than the number of fish to more appropriately weight the composition data relative to other data inputs

- Convert all composition data to conditional age-at-length to avoid a mixing of length compositions and age compositions being fit to. Using conditional age-at-length contains more detailed information about the relationship between size and age and provides a stronger ability to estimate growth parameters, especially the variance of size-at-age

Red Tide

- Enable rapid response sampling following severe events to quantify numbers, sizes and species composition of fish in fish kills
- Continue red tide index modeling efforts, specifically by ironing out issues with products derived from MODIS (e.g., de-band)
- Cooperative research with fishermen to track red tide blooms offshore and provide information on species composition, numbers and sizes in fish kills
- Evaluate impacts of red tides on food web dynamics and investigate recovery lags when forage base is impacted
- Conduct tagging studies to investigate response of Red Grouper and other species to red tide events, including fish movement and avoidance
- Use Vessel Monitoring System data to test hypotheses provided by stakeholders that vessels are shifting their distribution in response to fish moving during red tides
- Simulation test the various approaches for incorporating red tide mortality into the assessment model to determine the trade-offs associated with each approach

Projections

- Evaluate current approach used for setting up and conducting projections and consider conducting a meta-analysis of steepness to assist in set-up of projection